ABGs

Arterial Blood Gases

Metabolic Acidosis

$pH < 7.35$ $HCO_3 < 22$

Metabolic **E**qual

(pH + Bicarb / HCO_3^- move in equal / same directions)

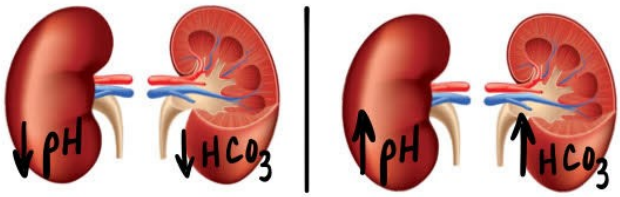
vs.

Metabolic Alkalosis

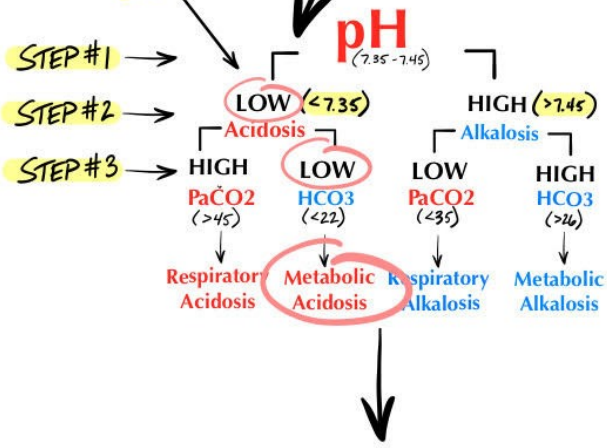
$pH > 7.45$ $HCO_3 > 26$

Metabolic **E**qual

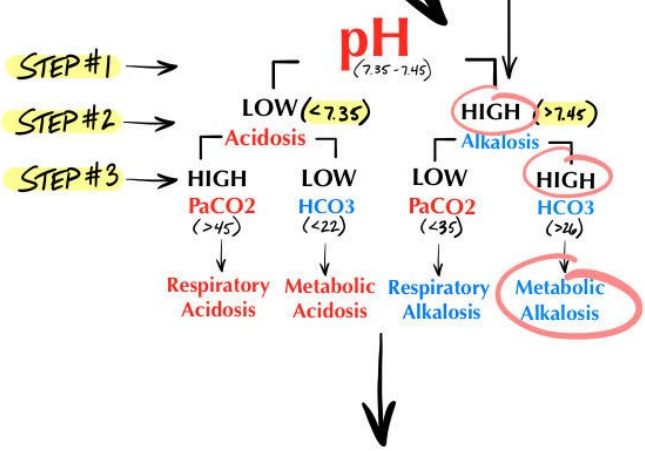
(pH + Bicarb / HCO_3^- move in equal / same directions)



Notice: Same Directions



Notice: Same Directions



Causes

- Acid Increase
  - Keto Acidosis
  - Starvation
  - Lactic Acidosis
    - Shock
    - Hypoxemia
- Decreased Bicarbonate (BASE)
  - Severe Diarrhea
  - Renal Failure

Signs/Symptoms

- Headache
- Lethargy
- Kussmaul's respirations
- Nausea
- Vomiting
- Confusion
- Coma

Treatments

- Improve Oxygenation
- Aimed at correcting underlying cause (DKA, shock, sepsis, diarrhea, renal disorders)

Causes

- Acid loss
  - Vomiting
  - Gastric suction → HCl loss
  - Diuretic use
    - K^+/Na^+ loss
- Antacids
- Hyperaldosteronism
  - Cushing's disease
  - Steroid use

Signs/Symptoms

- Weakness
- Muscle cramps
- Hyperactive reflexes
- Tetany
- Confusion
- Slow, shallow respirations to minimize CO_2 loss
- Nausea
- Seizures

Body needs this to counter alkalotic state

Treatments

- Aldosterone antagonists
- K^+ sparing diuretics
- Correct underlying cause